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Pre-Lab Questions

1. The BJT in Figure 1 is in a forward-active region with $V_{BE} = 0.7V$, $I_B = 12mA$, and $V_{CE} = 2.5V$.

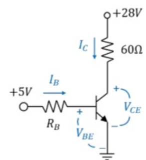


Figure 1: BJT in Common Emitter Configuration

   a) What is the power dissipated in the BJT?
   b) What is the power dissipated in the load?
   c) What is the power efficiency of the circuit?

2. The BJT in Figure 1 is in a saturation region with $V_{BE} = 0.8V$, $I_B = 30mA$, and $V_{CE} = 0.3V$.

   a) What is the power dissipated in the BJT?
   b) What is the power dissipated in the load?
   c) What is the power efficiency of the circuit?

a) $P = V \cdot I$

$= (2.5V)(425mA)$

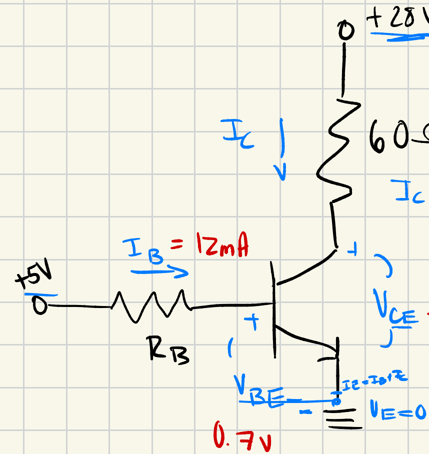
$= 1.0625\,W$

$\boxed{P = 1.06\,W}$

b) $P = I^2 \cdot R$

$= (0.425)^2 \cdot (60) = \boxed{10.8375\,W}$

1)



$I_E = I_B + I_C$

$I_E = \alpha I_C$

$I_C = \frac{28-2.5}{60}$

0.425 425mA

$\alpha = \frac{B}{B+1}$

$I_E = \frac{V_E}{R_E} = 0$

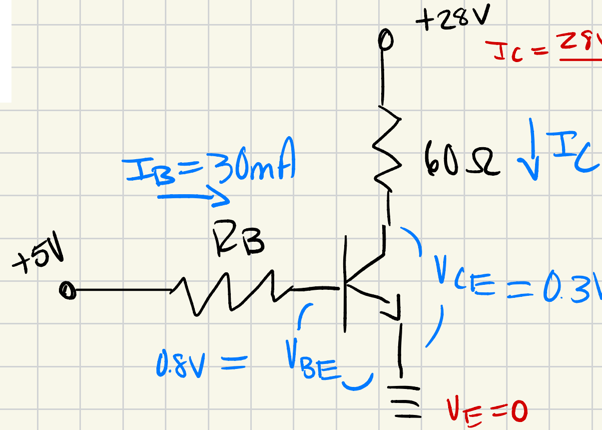
$R_B = \frac{5V - 0.7}{12mA} = 358\Omega$

c) $\% = \frac{10.8375W}{10.8375W + 1.06W} \times 100\%$

$= \boxed{91.09\%}$

2. The BJT in Figure 1 is in a saturation region with $V_{BE} = 0.8V$, $I_B = 30mA$, and $V_{CE} = 0.3V$.

a) What is the power dissipated in the BJT?
b) What is the power dissipated in the load?
c) What is the power efficiency of the circuit?



$$I_C = \frac{28V - 0.3V}{60\Omega} = 0.4616\ldots7$$

$$\Rightarrow 461mA$$

$$R_B = \frac{5V - 0.8V}{30mA} = 140\Omega$$

a) $P = V \cdot I$

$$= (0.3V)(461mA) = 0.1383$$

$$\boxed{P = 138.3mW}$$

b) $P = I^2 \cdot R$

$$= (0.461)^2 \cdot (60\Omega) = 12.75W$$

$$\boxed{P = 12.75W}$$

c) $$\% = \frac{12.75W}{12.75W + 0.1383W} \times 100\%$$

$$= \boxed{98.92\ \%}$$